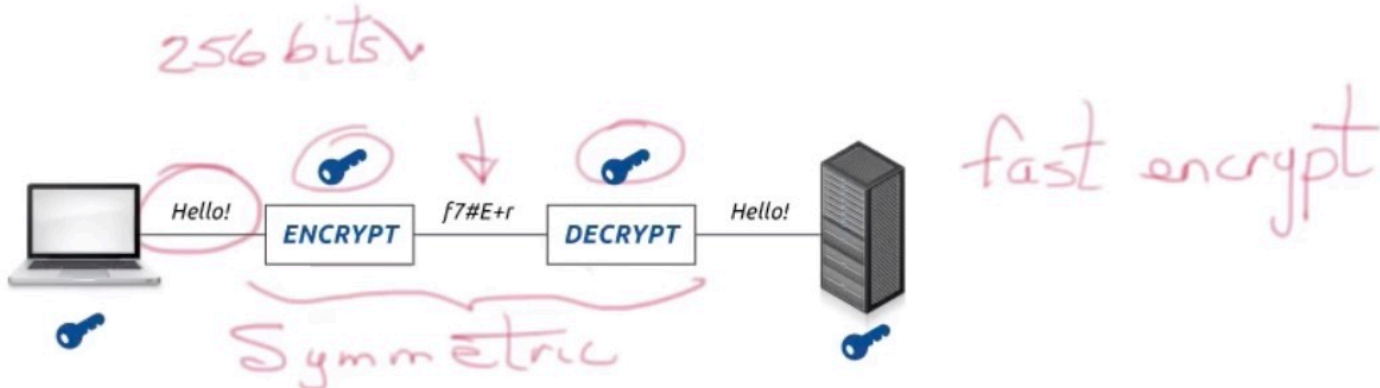
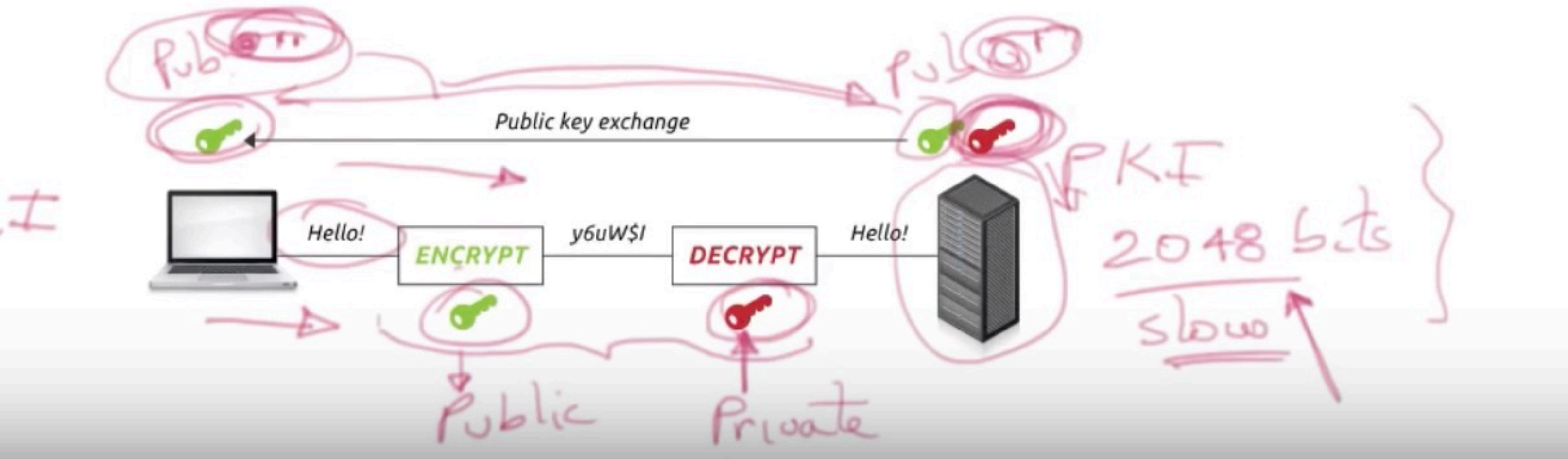
Review

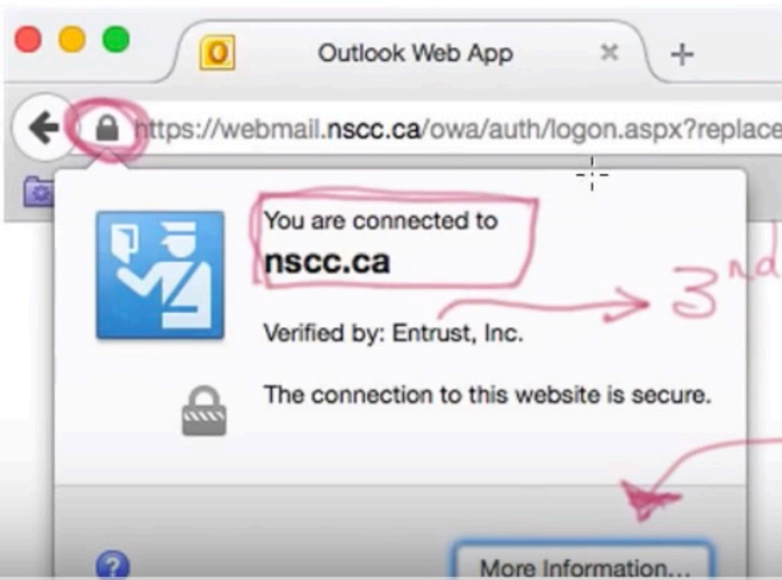

256 bits
Hello!
ENCRYPT
f7#E+r
DECRYPT
Hello!
fast encrypt
Symmetric


Pub
Pub
Public key exchange
PKI
Hello!
ENCRYPT
y6uW$I
DECRYPT
Hello!
2048 bits
slow
Public
Private

Our Example

Secure Sockets Layer.

Using a Certificate for Web SSL communications



3rd Party is validate

Trust

What is a digital certificate?

electronic credentials used to assert the online identities of individuals, computers, and other entities on a network

similar to ID cards (passports, driver licenses)

contain public key & identify of owner

Certificate Issuer

issued by a certificate authority (CA) that validate the identify of the certificate holder

self-signed certificate

Server

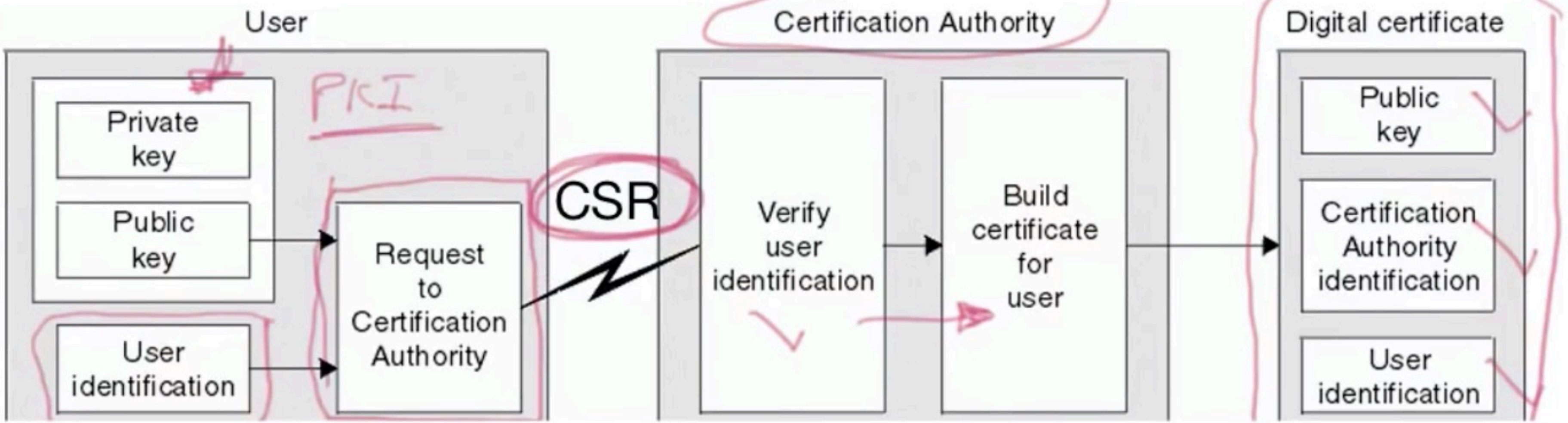
Web Site

Certificate

Trust

Certificate Signing Request

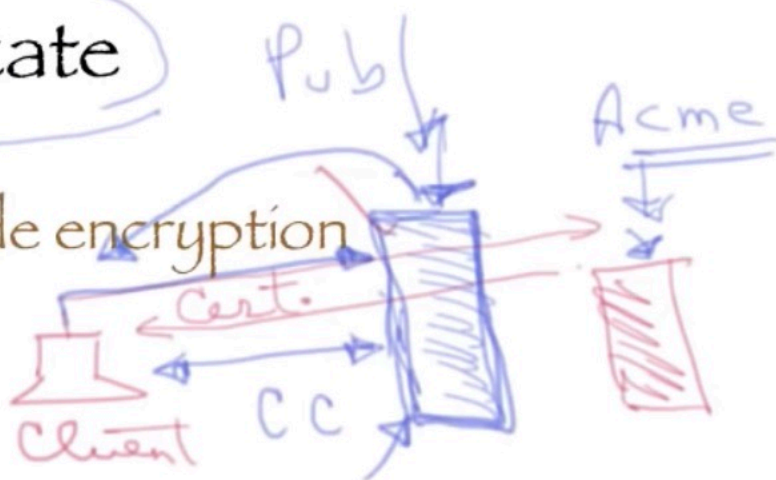
Apache



This command creates a 2048-bit private key (`domain.key`) and a CSR (`domain.csr`) from scratch: cc.CA

```
openssl req \
       -newkey rsa:2048 -nodes -keyout domain.key \
       -out domain.csr
```

Self-Signed Certificate

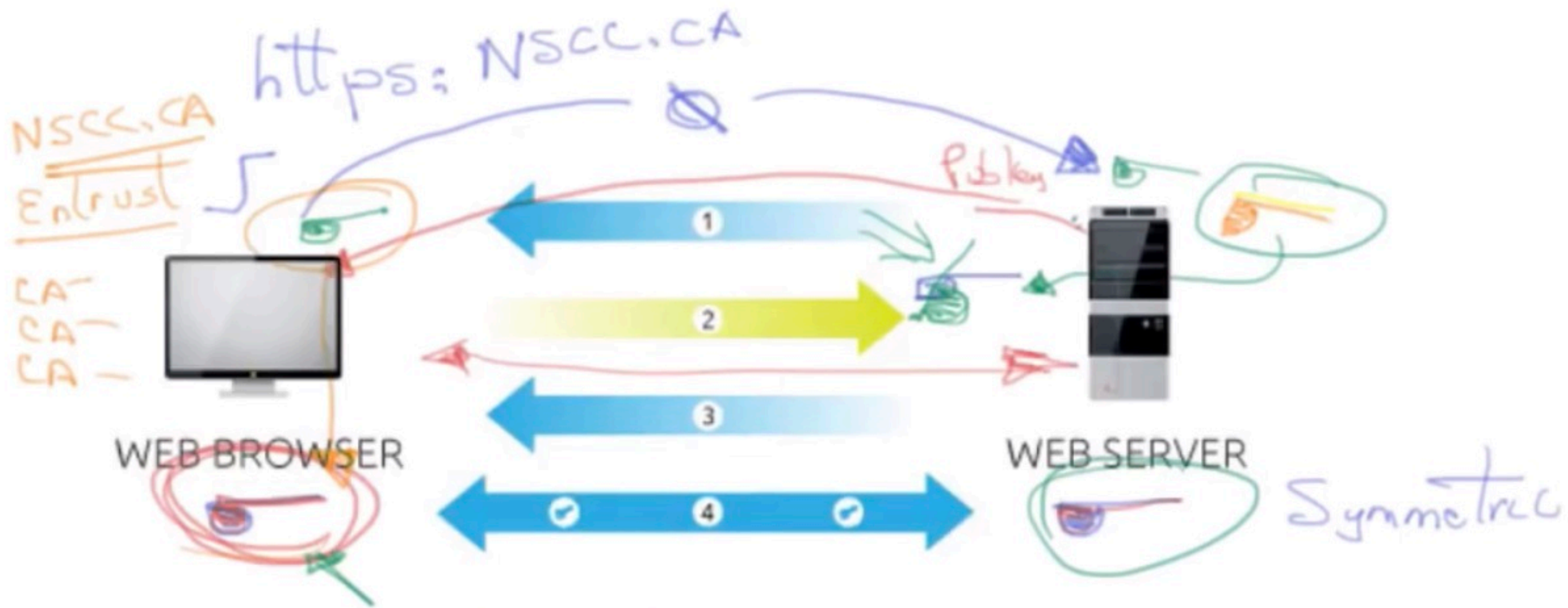


- SSL Certificates only provide encryption
- trust is why you use a CA

man in the middle attack

set up a CA (self-signed) and create a root-certificate

Site NOT Trusted



1. **Server** sends a copy of its asymmetric public key.

2. **Browser** creates a symmetric session key and encrypts it with the server's asymmetric public key.

3. **Server** decrypts the asymmetric public key with its asymmetric private key to get the symmetric session key.

4. **Server** and **Browser** now encrypt and decrypt all transmitted data with the symmetric session key. This allows for a secure channel because only the browser and the server know the symmetric session key, and the session key is only used for that session. If the browser was to connect to the same server the next day, a new session key would be created.

Root CA

↓

Intermediate CA

NSCC.CA → CSR

ICA

SSL Cert
ICA

ICA

Layers of security

Keys → inaccessible

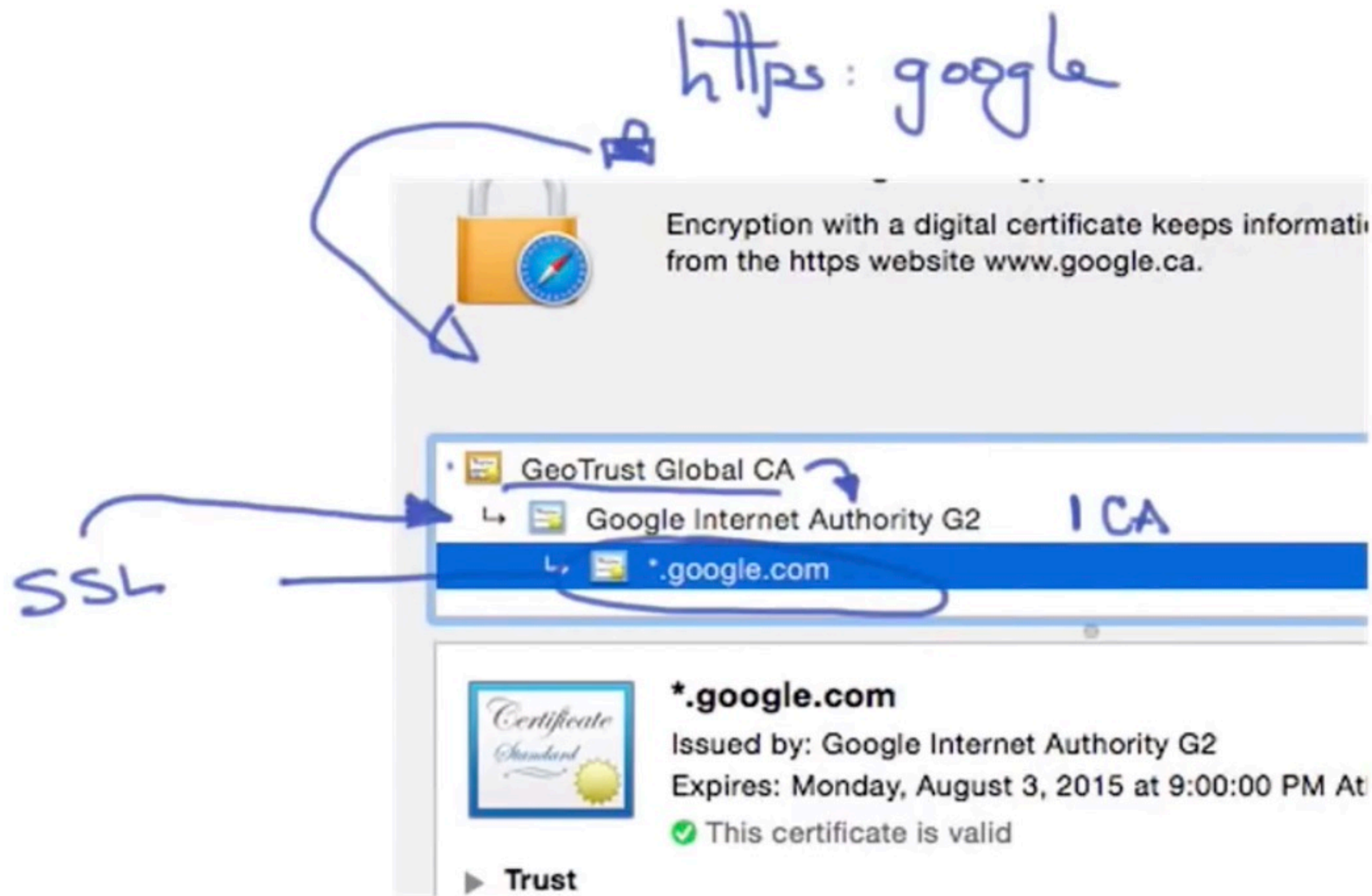
SSL → NSCC.CA

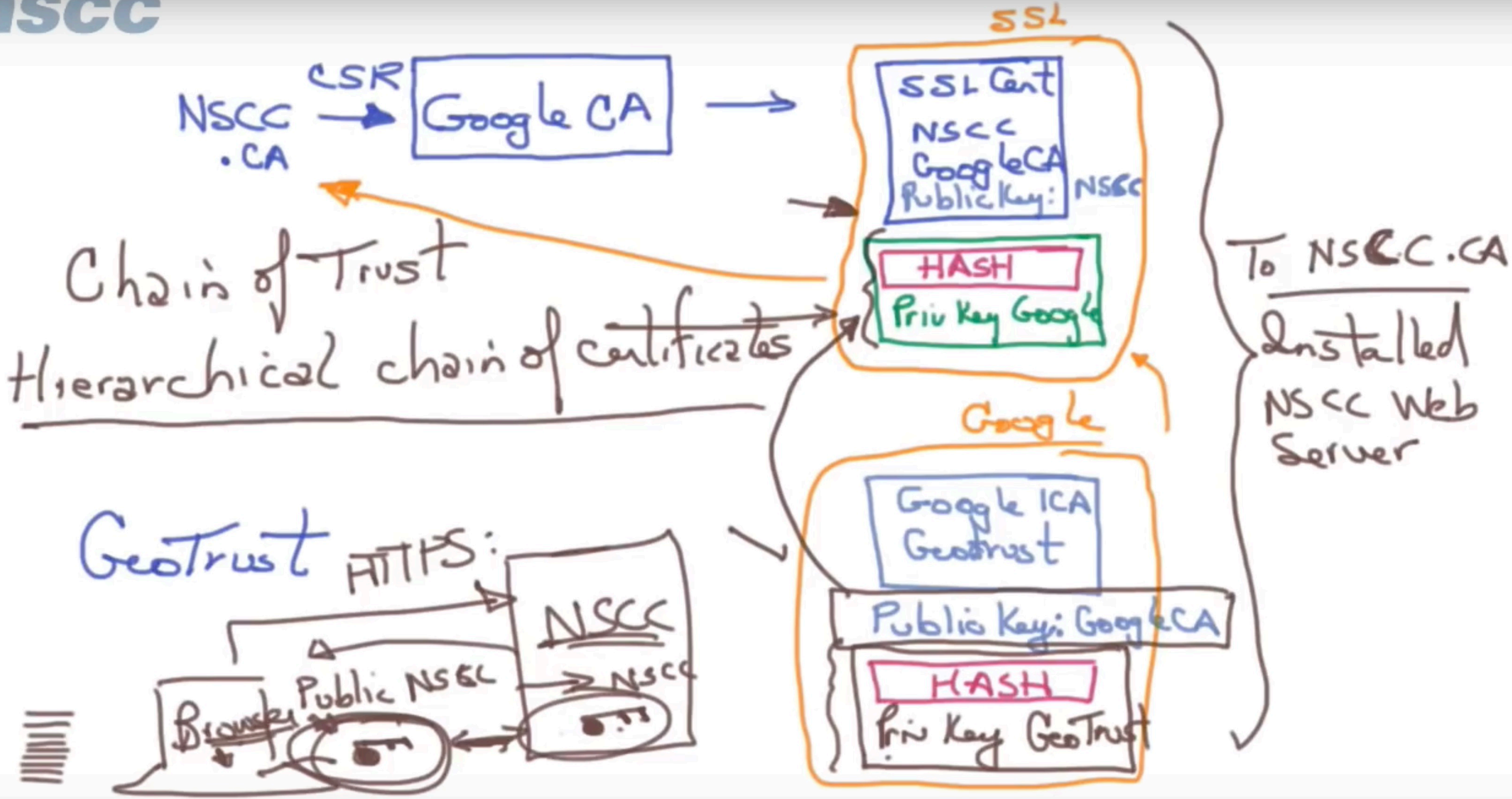
↓

ICA

↓

CA


https: google
Encryption with a digital certificate keeps informati
from the https website www.google.ca.
GeoTrust Global CA
Google Internet Authority G2
1 CA
SSL
*.google.com
Certificate
Standard
*.google.com
Issued by: Google Internet Authority G2
Expires: Monday, August 3, 2015 at 9:00:00 PM At
This certificate is valid
Trust


CSR
NSCC .CA
Google CA
SSL
SSL Cert
NSCC
GoogleCA
Public Key: NSCC
HASH
Priv Key Google
Chain of Trust
Hierarchical chain of certificates
To NSCC.CA
Installed
NSCC Web Server
Google
Google ICA
GeoTrust
Public Key: GoogleCA
HASH
Priv Key GeoTrust
GeoTrust
HTTPS:
NSCC
Public NSCC
NSCC
Browser